Status Report #3

Nov 12, 2026

The text detector passed its final test on unseen data and now runs in the RealEye backend, replacing the 50% placeholder. The first image detector meets both project targets on validation data, and the extension can now scan visible content and mark results on the page.

Reporting period: October 30 – November 12, 2026.

Progress

The team worked in three parallel tracks: machine learning, the Chrome extension, and the server.

Machine learning (Tara and Noa)

Area	What we did	Status
Text: error analysis	Found why the first model flagged 41% of human writing as AI: unevenly mixed training data and overconfident scores. Fixed both and retrained.	Done
Text: calibration	Replaced the 0.999 threshold with calibrated scores, so the percentages shown to users are trustworthy.	Done
Text: final test	Tested on **51,002 texts** the model had never seen: **F1 0.89** (target ≥ 0.75) and **5.9% false-positive rate** (target ≤ 10%).	Targets met
Text: backend	Connected the trained text model to the RealEye backend and made the extension's result wording more cautious.	Done
Image: model and dataset	Selected **CLIP** with a small trained classifier on top, and the **OpenFake** dataset of real and AI-generated images.	Selected

Area	What we did	Status
Image: data preparation	Prepared **45,000 images**, processing real and AI images identically so the model can't rely on file details like format or size.	Done
Image: first model	Validation results: **F1 0.80 at a 10% false-positive rate**.	Validation targets met
Image: error analysis	Most false alarms are polished or digitally made real images (posters, CGI, studio photos); very realistic AI photos are hardest to catch. One fix (balancing image styles) was tested and did not help.	In progress

Chrome extension (Charlize)

Area	What we did	Status
Popup redesign	Redesigned the popup's layout, branding and accessibility, with clearer loading, error and retry states.	Done
Page scanning	Improved text extraction to skip menus, footers and hidden or duplicate text; added visible-area and selected-content scans, cancelling, and batching for scans over 10 items.	Done, testing in Chrome
Inline warnings	Added markers and warning labels shown directly on the webpage.	First version
Testing	Automated tests and production builds pass; some Chrome runtime errors are still being fixed.	In progress

Server (Anastasia and Anastasia)

Area	What we did	Status
Request validation	Set up the backend locally and reviewed the input rules: request size limits, required fields, and supported image formats and sizes. Started improving error handling.	In progress
Security and detection	Planned rate limiting (60 requests per minute), C2PA and SynthID checks, and timeouts so a slow check can't block results.	Planned

Next steps

The next period focuses on finishing the image detector and making full scans reliable from the extension to the server.

1. **Machine learning:** decide how to reduce image false alarms, calibrate the image model, run its one-time final test, and connect it to the backend.
2. **Extension:** fix the remaining Chrome errors, finish the inline warning labels, and add image scanning on webpages.
3. **Server:** finish request and image validation, add rate limiting, and start the C2PA check.
4. **Whole team:** test complete scans on real webpages with both trained models.

Troubleshooting and open questions

Image false alarms: the image model wrongly flags many polished or digitally made real images, such as posters, CGI and retouched studio photos. Should we accept this as a documented limitation, retrain with more weight on these images, or try a larger model?

Unfamiliar writing: the text model is less reliable on writing styles it never trained on, such as news articles and dialogue. We will note this as a known limitation.

Chrome scanning errors: the newest scanning features pass automated tests, but some pages still fail to scan in Chrome. These need debugging before end-to-end testing.

SynthID access: we still need to confirm whether an authorized SynthID detector is available; until then the server reports it as unavailable, which the SRS allows.